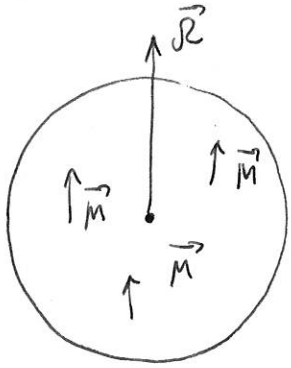
Rotating magnetic sphere



$$\vec{D} = \varepsilon\vec{E} - \frac{\vec{V}\times\vec{H}}{c^2} + \varepsilon\,\vec{V}\times\vec{B}$$

$$\vec{B} = \mu\vec{H} + \frac{\vec{V}\times\vec{E}}{c^2} - \mu\,\vec{V}\times\vec{D}$$

To the zeroth (in $\frac{V}{c}$) order

$$\vec{B} = \mu\vec{H} = \mu_0(\vec{H}+\vec{M})$$

$$\nabla\cdot\vec{B} = 0 \qquad \nabla\times\vec{H} = 0$$

Due to second equation

$$\vec{H} = \nabla\psi$$

Due to the first equation $\nabla^2\psi = 0$.

Inside the sphere we expect a uniform field

$$\psi_{in} = \vec{H}_{in}\cdot\vec{r} = H_{in}\cos\theta$$

outside we expect a dipole field

$$\psi_{out} = \frac{\vec{A}\cdot\vec{r}}{r^3} = \frac{A}{r^2}\cos\theta$$

Boundary conditions:

1) $\vec{H}_{\parallel}$ is contin. $\longrightarrow$ ψ_i is contin.

$$\boxed{H_{in} = \frac{A}{R^3}}$$

2) $B_n = \mu_0(\vec{H}+\vec{M})_n$ is contin.

$$\mu_0(H_{in}\cos\theta + M\cos\theta) = -\frac{2A}{R^3}\cos\theta\,\mu_0$$

or

$$\boxed{H_{in} + M = -\frac{2A}{R^3}}$$

$$A = -\frac{MR^3}{3} \qquad H_{in} = -\frac{M}{3}$$

or in vector form:

$$\boxed{\vec{H}_{in} = -\frac{1}{3}\vec{M} \qquad \vec{B}_{in} = \frac{2\mu_0}{3}\vec{M}}$$

We now use these expressions in the expression for $\vec{D}$:

$$\vec{D} = \varepsilon\vec{E} + \varepsilon\frac{2\mu_0}{3}\vec{v}\times\vec{M} + \frac{\vec{v}\times\vec{M}}{c^2} = \varepsilon\vec{E} + \frac{\mu_0(2\varepsilon+\varepsilon_0)}{3}\vec{v}\times\vec{M}$$

Taking now $\nabla\cdot$ of both sides and using

$$\nabla\cdot\vec{D} = 0 \quad \text{and} \quad \nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0}$$

we find the density of (bulk) bound charges:

$$\rho = -\frac{\varepsilon_0\mu_0(2\varepsilon+\varepsilon_0)}{3\varepsilon}\nabla\cdot(\vec{v}\times\vec{M})$$

since $\vec{M} = \text{const}$

$$\nabla\cdot(\vec{v}\times\vec{M}) = \vec{M}\cdot(\nabla\times\vec{v}) = \vec{M}\cdot\nabla\times(\vec{\omega}\times\vec{r})$$

Similarly,

$$\nabla\times(\vec{\omega}\times\vec{r}) = \vec{\omega}(\nabla\cdot\vec{r}) - (\vec{\omega}\cdot\nabla)\vec{r} =$$

$$= 3\vec{\omega} - \omega\frac{\partial}{\partial z}\vec{r} = 3\vec{\omega} - \omega\hat{z} = 2\vec{\omega}$$

so that $\nabla\cdot(\vec{v}\times\vec{M}) = 2\vec{\omega}\cdot\vec{M} = 2\omega M$

and

$$\rho_{bulk} = -\frac{2\varepsilon_0\mu_0(2\varepsilon+\varepsilon_0)}{3\varepsilon}\omega M = -\frac{2(2\varepsilon+\varepsilon_0)}{3\varepsilon c^2}\omega M$$

This charge should be compensated by the surface charge which can be found by the explicit solution of electrostatics equations. The solution is greatly simplified when

$$\varepsilon \approx \varepsilon_0$$

Then

$$\vec{D} - \varepsilon_0 \vec{E} = \mu_0 \varepsilon_0 \vec{V} \times \vec{M} = \frac{1}{c^2}(\vec{\omega} \times \vec{r}) \times \vec{M} =$$

$$= \frac{1}{c^2}\left(\vec{r}\,\omega M - \vec{\omega}(\vec{r} \cdot \vec{M})\right)$$

However, $\vec{D} - \varepsilon_0 \vec{E} = \vec{P}$ and P_n at $r = R$ gives surface charge density:

$$\sigma_{sur} = P_n = \frac{1}{c^2}\left(R\,\omega M - \omega M R \cos^2\theta\right) =$$

$$\boxed{\sigma_{sur} = \frac{\omega M R}{c^2} \sin^2\theta}$$

at the same time for $\varepsilon = \varepsilon_0$ from the general expression on page 1 bulk charge density is

$$\boxed{\rho_{bulk} = -\frac{2\omega M}{c^2}}$$